

Biomedical Image Analysis using a Quantum Computer



iQuCodeFest 2026

<https://github.com/QuantumETS/iQuCodeFest-TeamA-Submission>

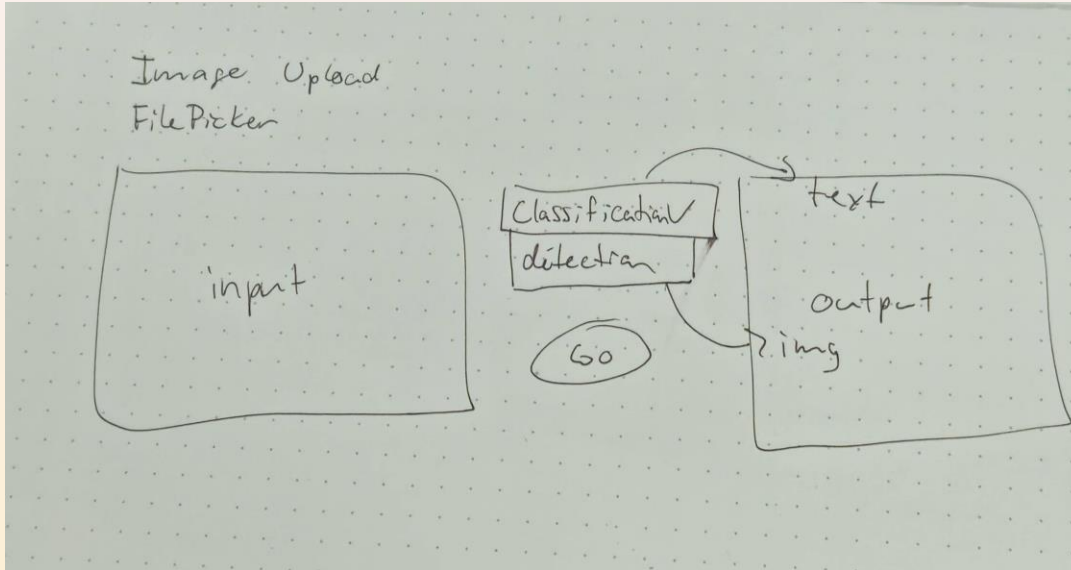
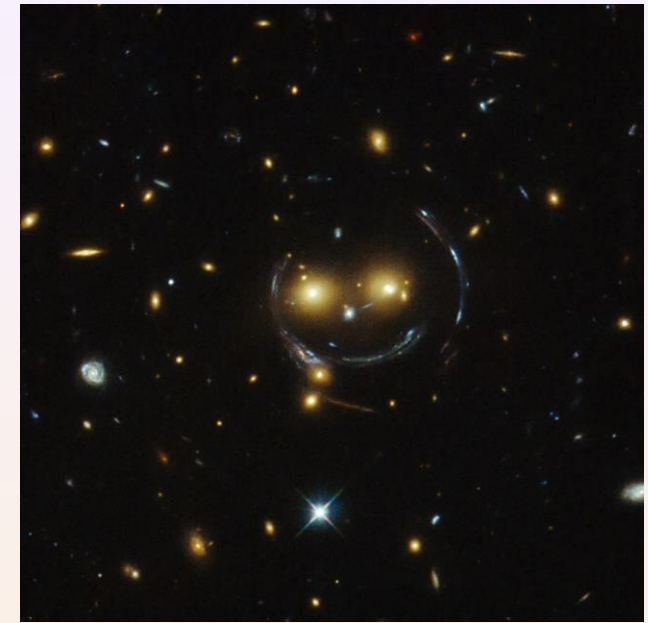
Samuel Richard
Guy-Philippe Nadon
Jacob Bernier
Victor Rochon

Quantum ÉTS
Team A

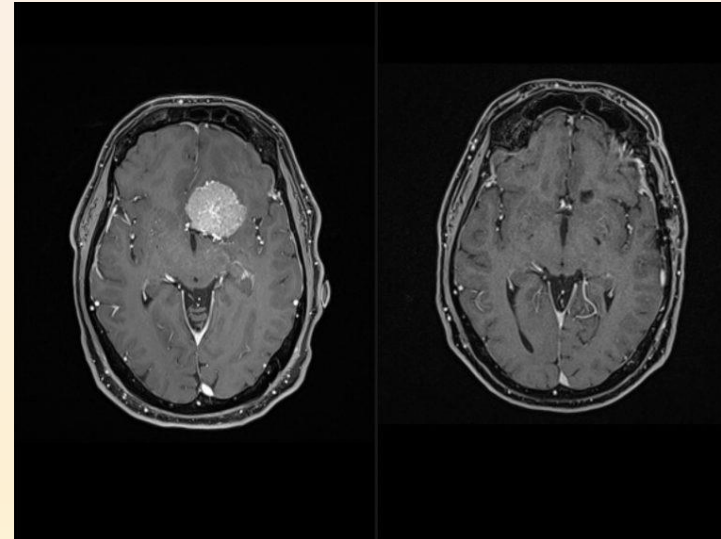
Aiming at helping science

Problems of interest for image analysis, especially segmentation !

- Finding gravitationnal lenses
- Detecting illness in medical images

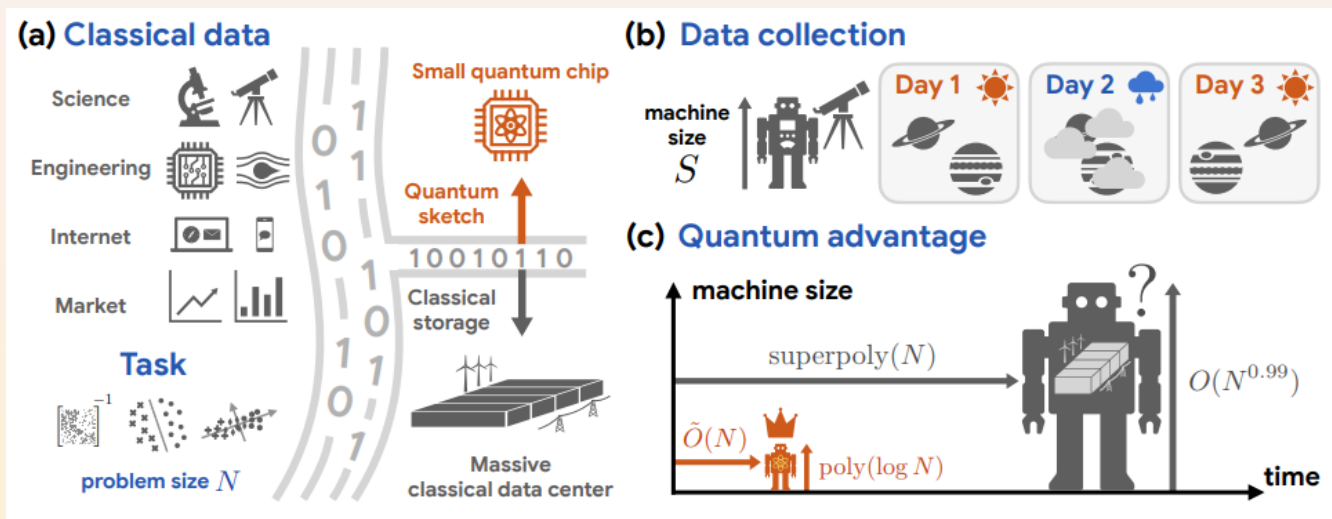


Sketch of our software interface

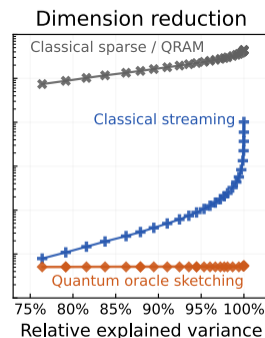
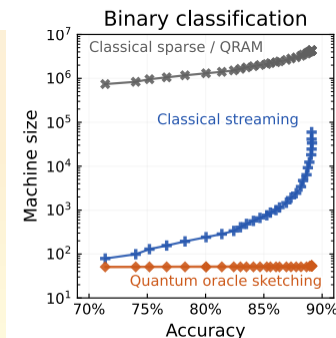
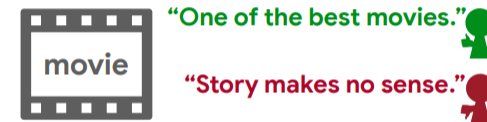


Preliminary exploration with Quantum Sketching

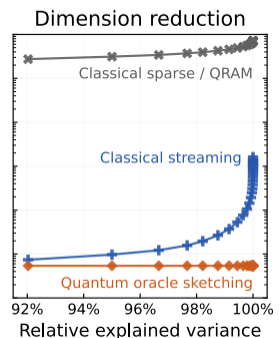
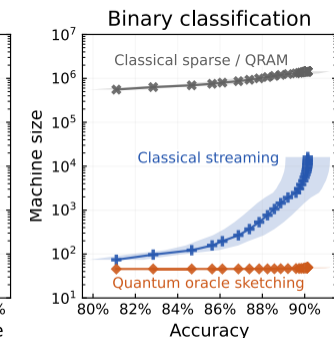
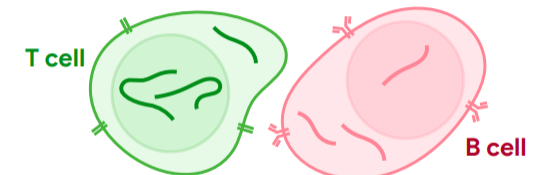
- <https://github.com/haimengzhao/quantum-oracle-sketching/tree/main>
- <https://arxiv.org/pdf/2604.07639>



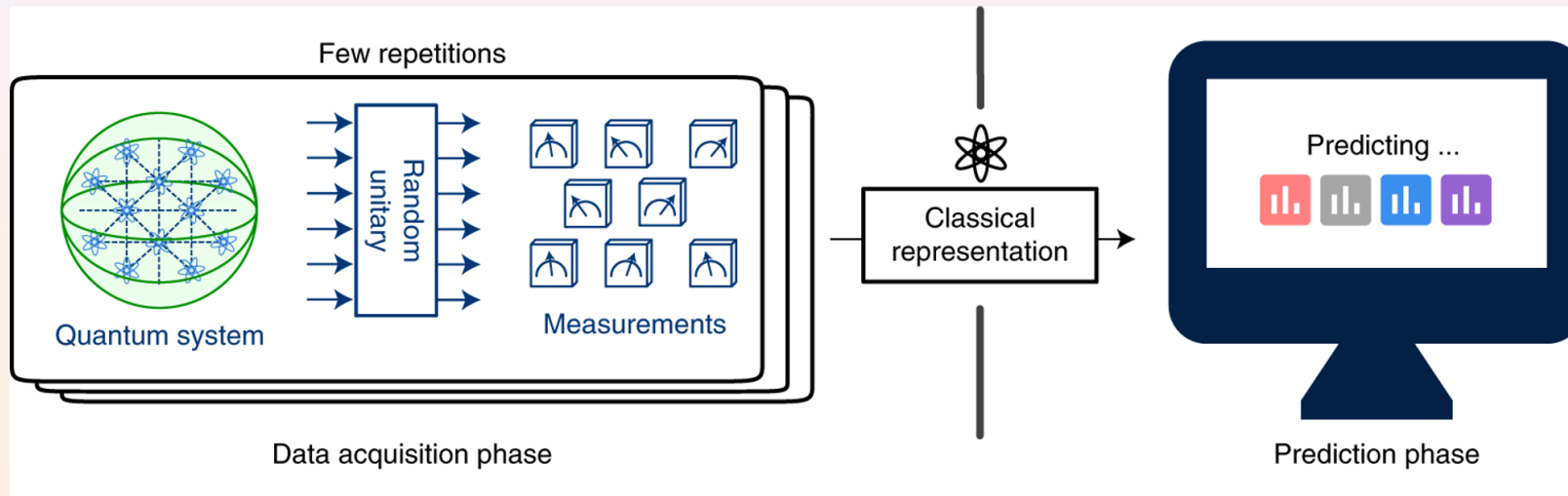
Movie review sentiment analysis



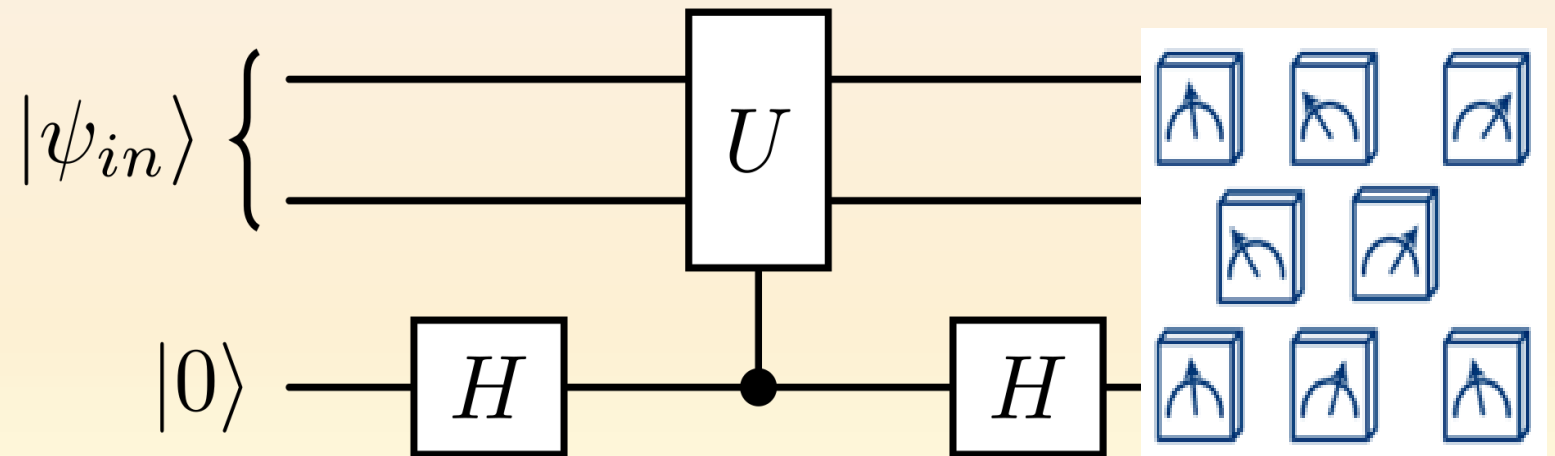
(b) Single cell RNA sequencing



Interferometric Classical Shadow



From https://pennylane.ai/demos/tutorial_classical_shadows

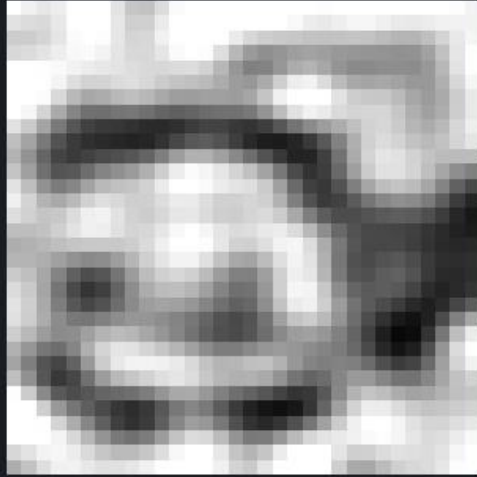


Quantum Sketching $k = 43$, with top coefficient

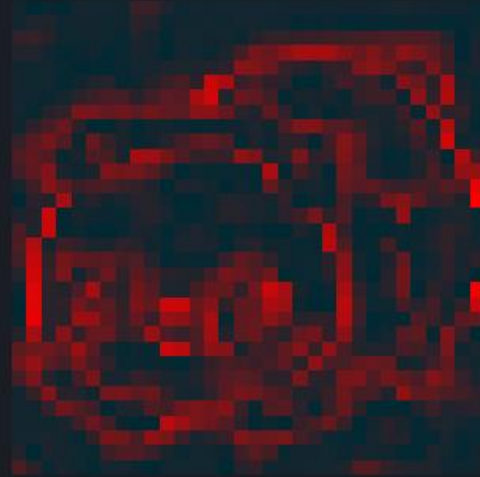
Input image



Decoded reconstruction



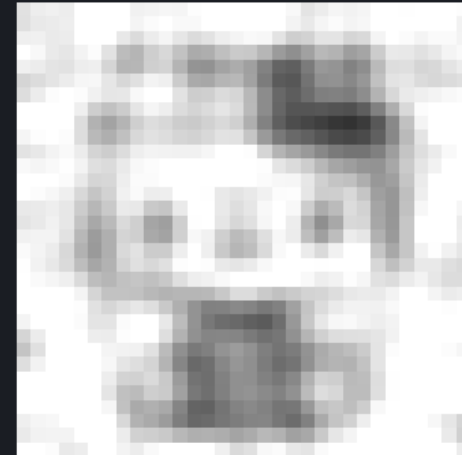
Absolute error



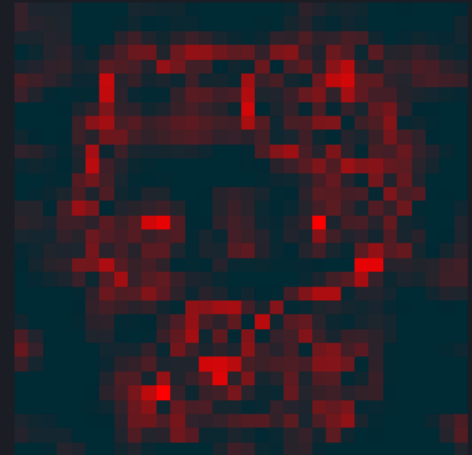
Input image



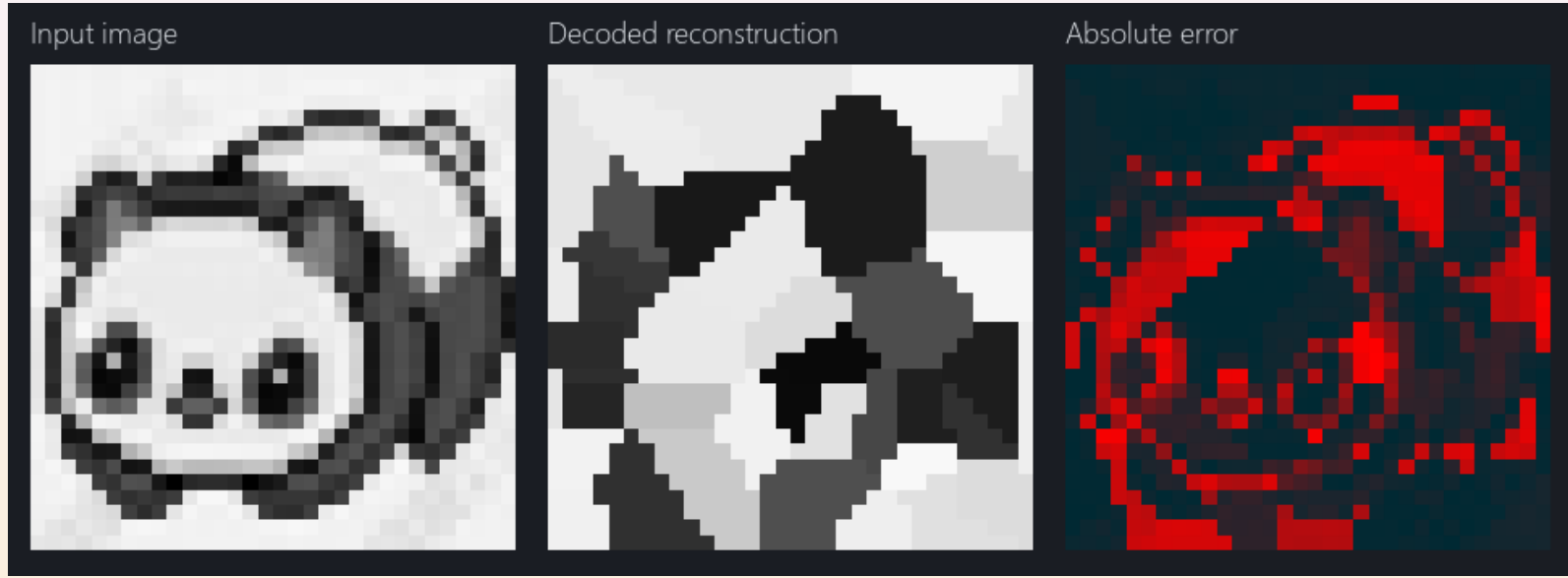
Decoded reconstruction



Absolute error

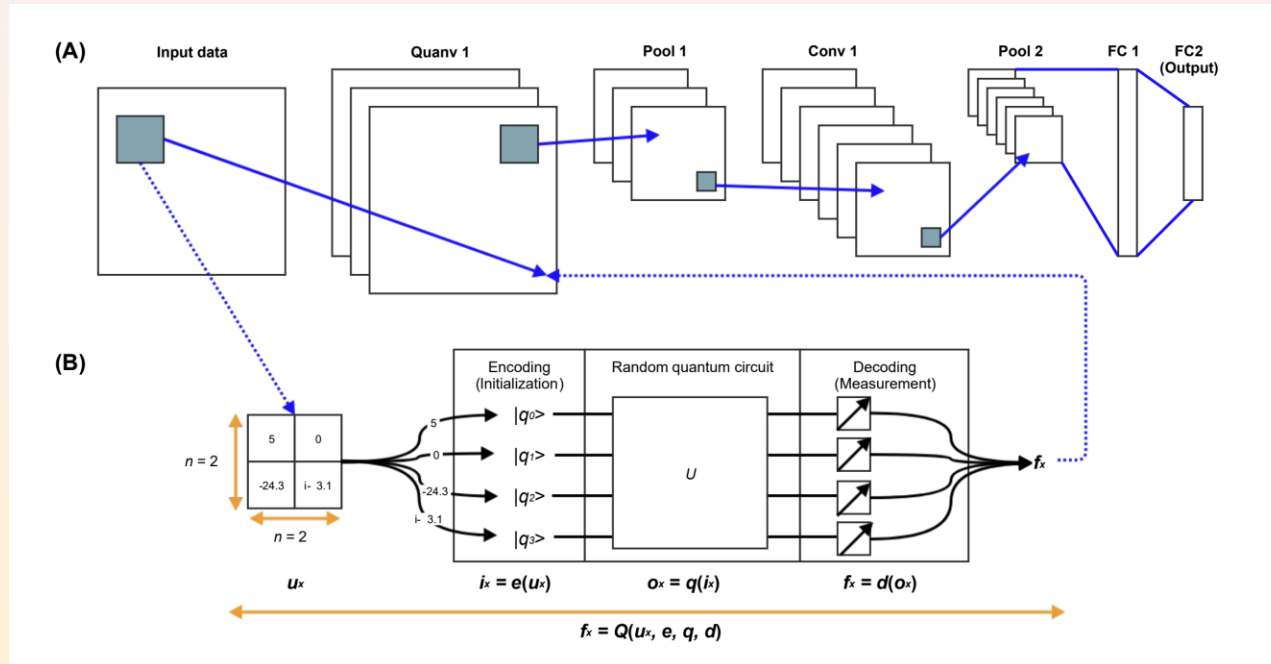


Quantum Sketching $k = 43$, random

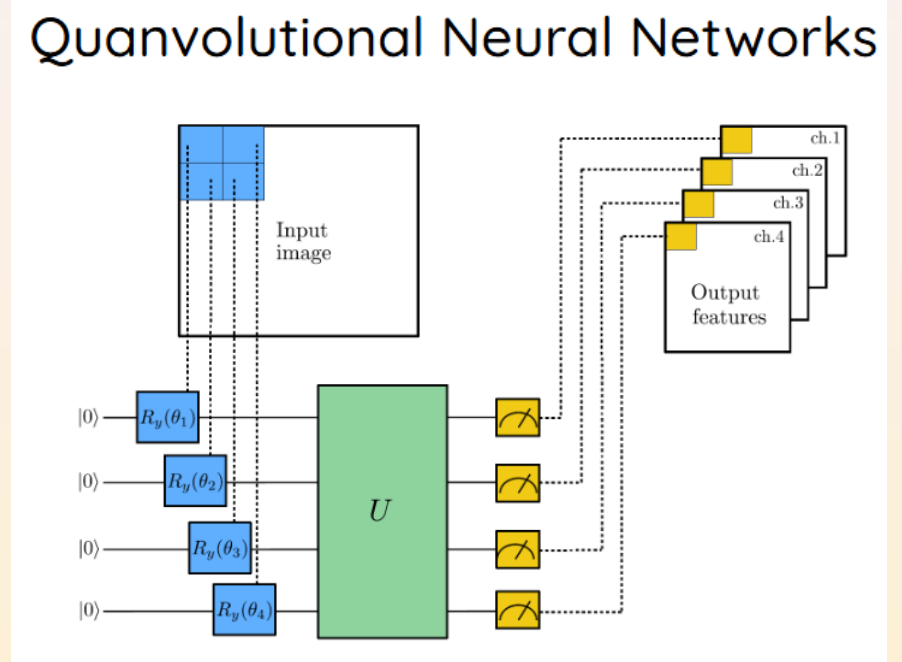


Redirection towards QCNNs

Quantum Convolutional (Quanvolutional) Neural Networks



<https://arxiv.org/pdf/1904.04767>



https://pennylane.ai/demos/tutorial_quanvolution

Choice of dataset



THOMAS · UPDATED 3 YEARS AGO

Brain tumors 256x256

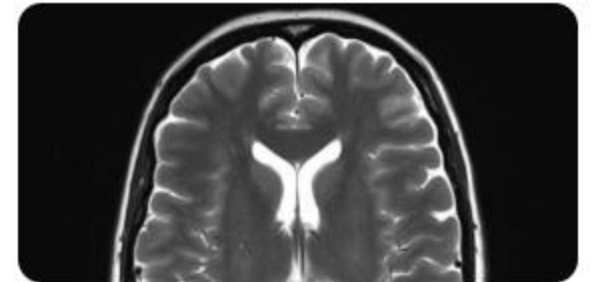
A Refined Brain Tumor Image Dataset with Grayscale Normalization and Zoom

kaggle

▲ 84

<> Code

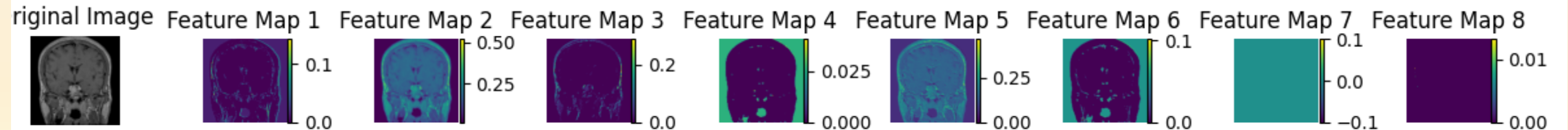
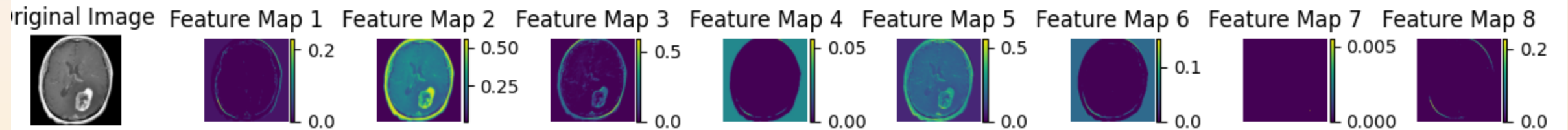
Download



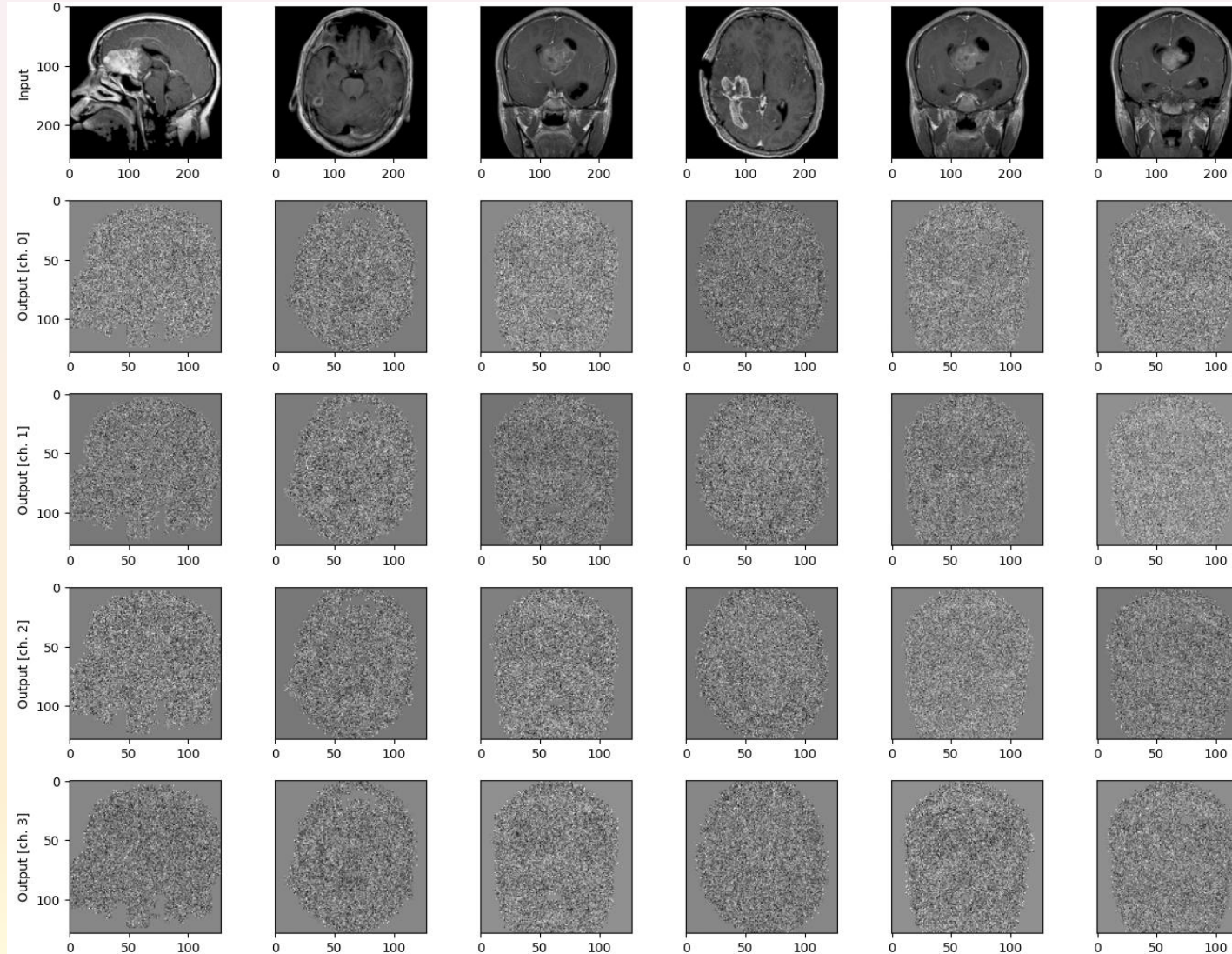
Quantum Feature Extraction

Classical Result – Convolutional neural network simulation

Each feature map represent the individual value given by the 4x4 kernel

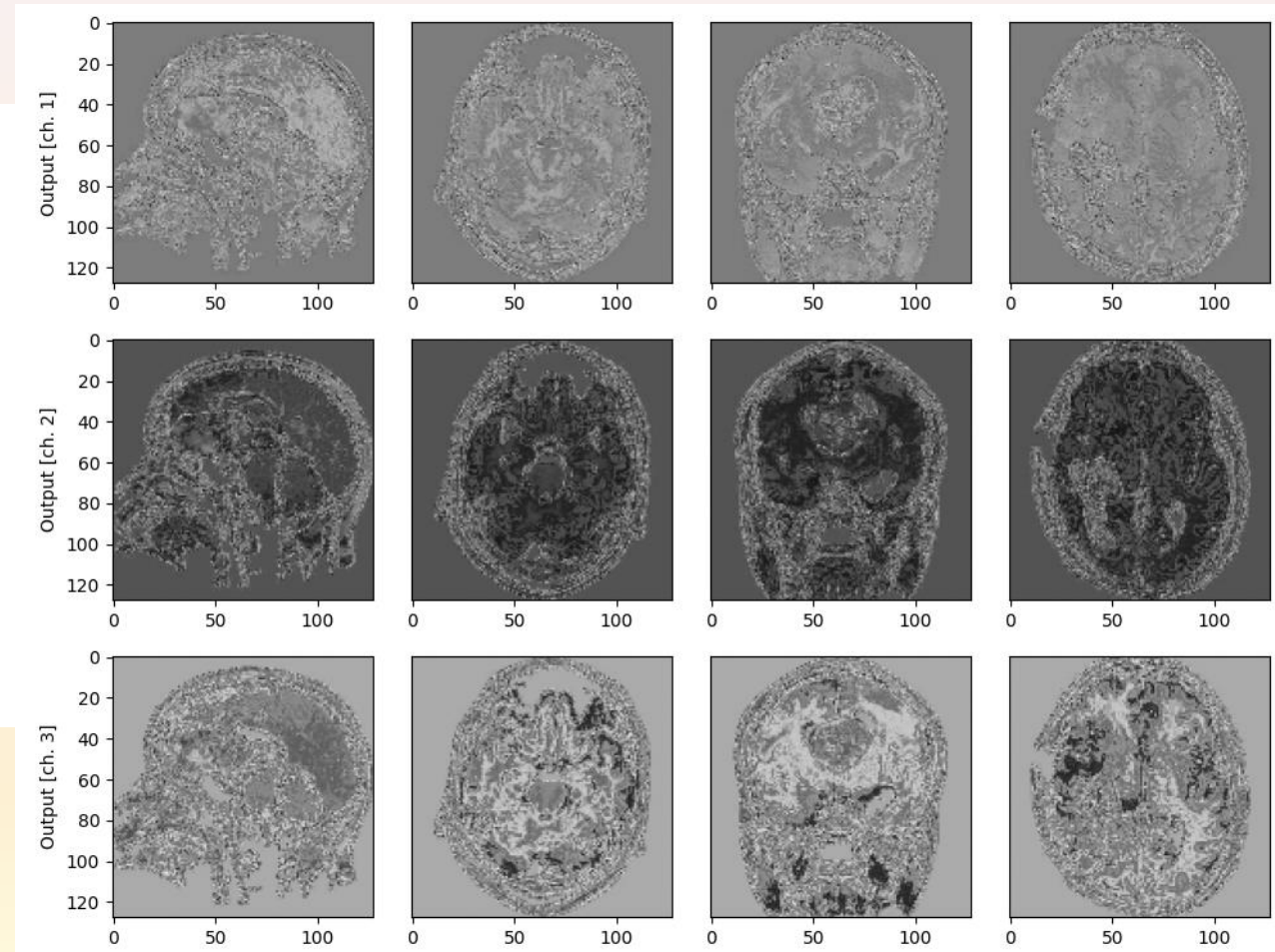
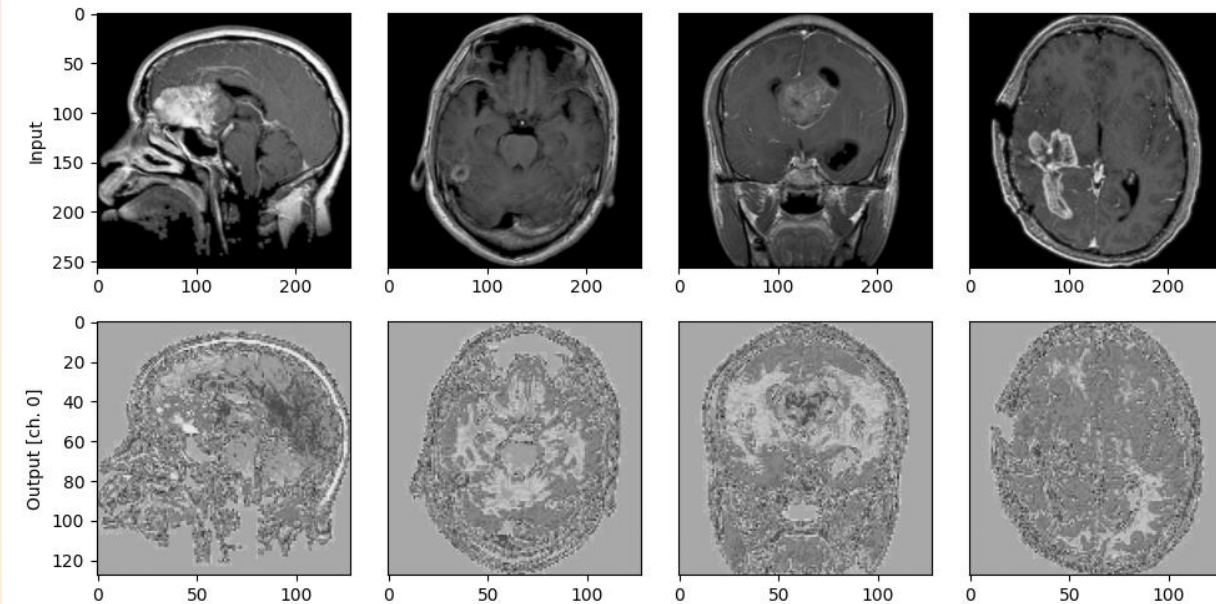


Quantum simulation result – Quanvolutional neural networks – failure

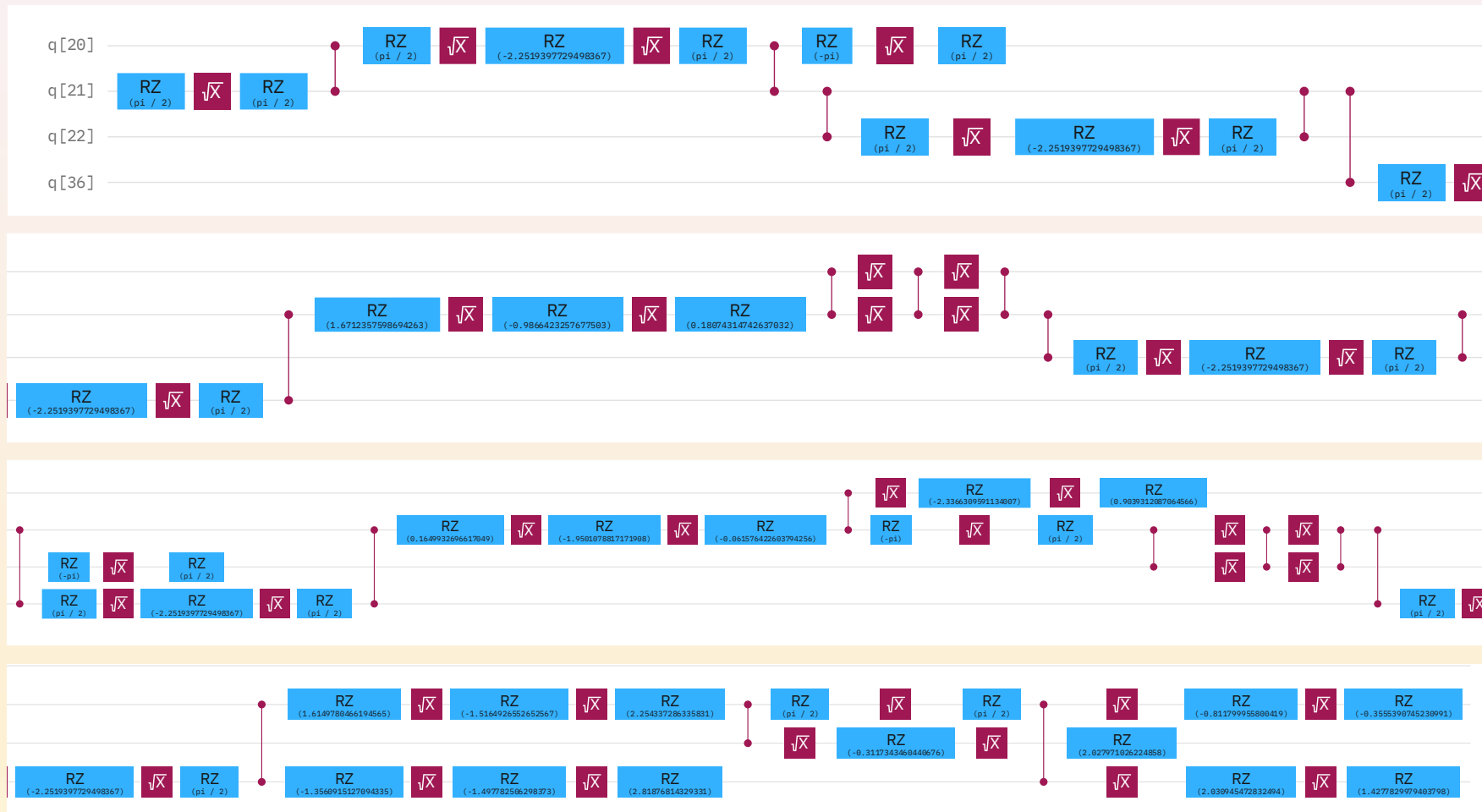


Quantum simulation result – Quanyvolutional neural networks – success!

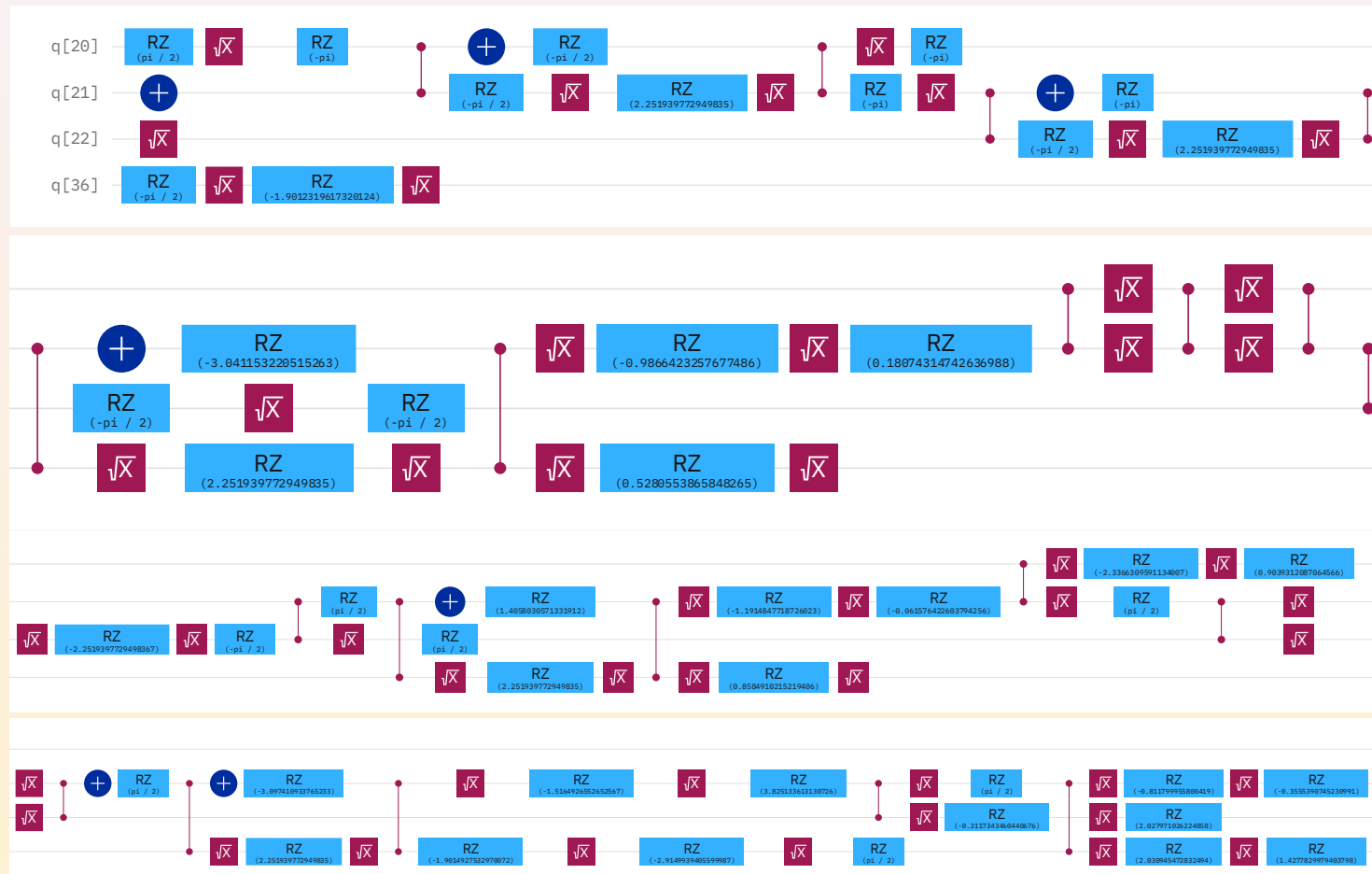
1 qubits = 1 feature map. Each row is a different feature map



Quantum convolutional neural networks – Hardware run Circuit non-optimized



Quantum evolutionary neural networks – Hardware run Circuit optimized

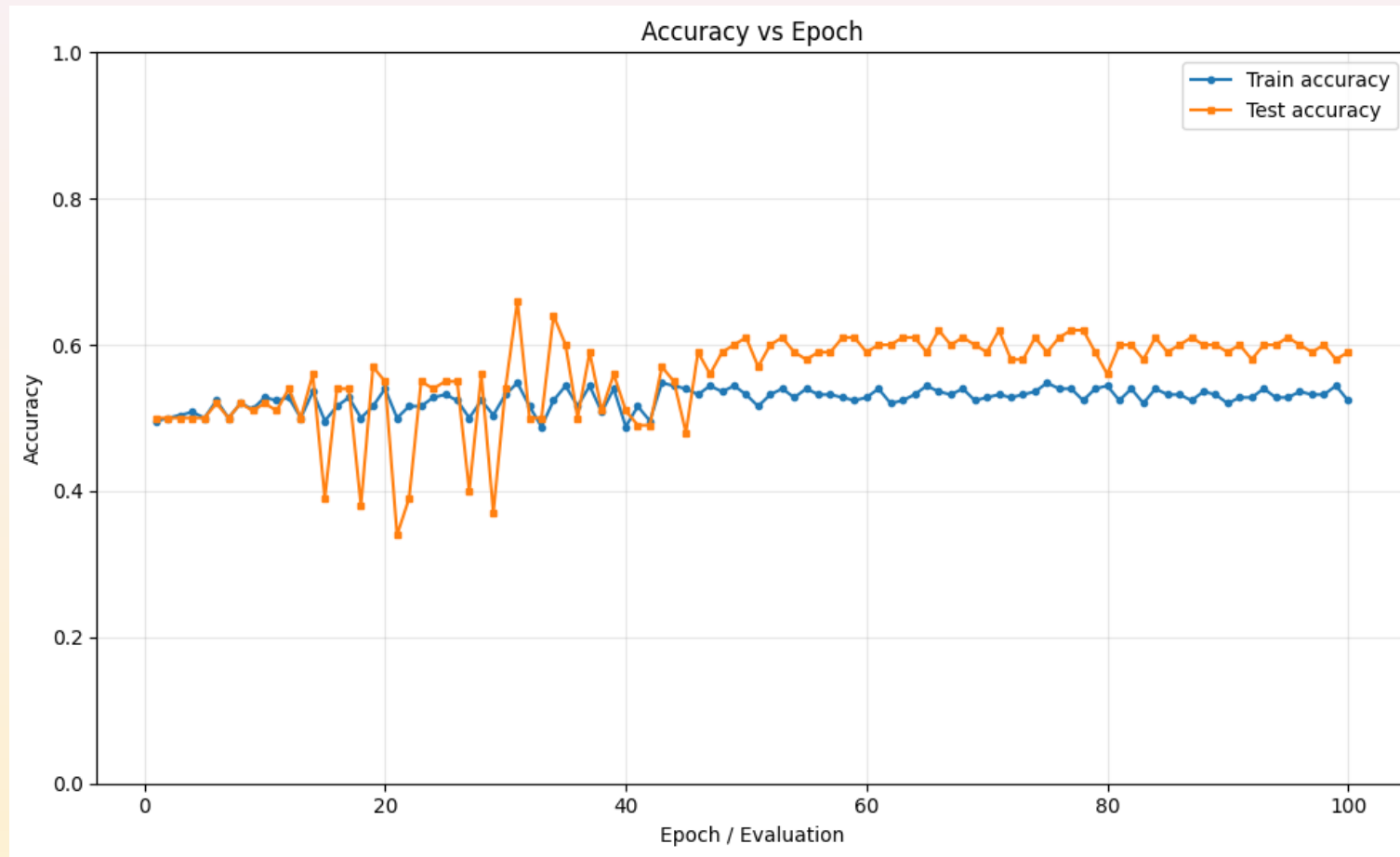


Still waiting on the results!

User Guy-Philippe Nadon	Region Washington DC (us-east)	Status In progress	Created: Jun 2, 2026, 11:07 AM
Instance Hackathon IQUCodeFest Juin 2026	Mode Session	Status reason session_activated	Pending: 3m 45s
Quantum computer ibm_quebec	Tags	Total usage -	In progress: Jun 2, 2026, 11:11 AM
			Completed
		- optimization_level = 3	
		- resilience_level = 2	
		- measurement mitigation	
		- dynamical decoupling with XY4	
		- ZNE with gate folding and [1, 3, 5] noise factors	
		- linear + exponential extrapolators	
		- gate and measurement twirling	
		- 4096 shots	
Jobs		Tags	
☐	ID	Status	Created
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☐	d8ffeco7jphs73...	Completed	6/2/26, 11:30 AM
☐	d8ffdl1vjngc73a...	Completed	6/2/26, 11:29 AM
☐	d8ffct1vjngc73a...	Completed	6/2/26, 11:27 AM
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☐	d8ffalpvjngc73a...	Completed	6/2/26, 11:23 AM
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☐	d8ff95bo3njc73...	Completed	6/2/26, 11:19 AM
☐	d8ff8dbalsvc73...	Completed	6/2/26, 11:18 AM

Quantum Classifier of Medical Imagery

Training in a time crunch...



Quantumvolutional neural network classification quantum pipeline

